

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1  
library(tidyverse); library(here); library(rvest)  
getwd(); here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
LWSP_site <- read_html(
  'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024'
)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
system <- LWSP_site %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()

PWSID <- LWSP_site %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()

ownership <- LWSP_site %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()

MaxDayUse <- LWSP_site %>%
  html_nodes("th~ td+ td") %>%
  html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc...

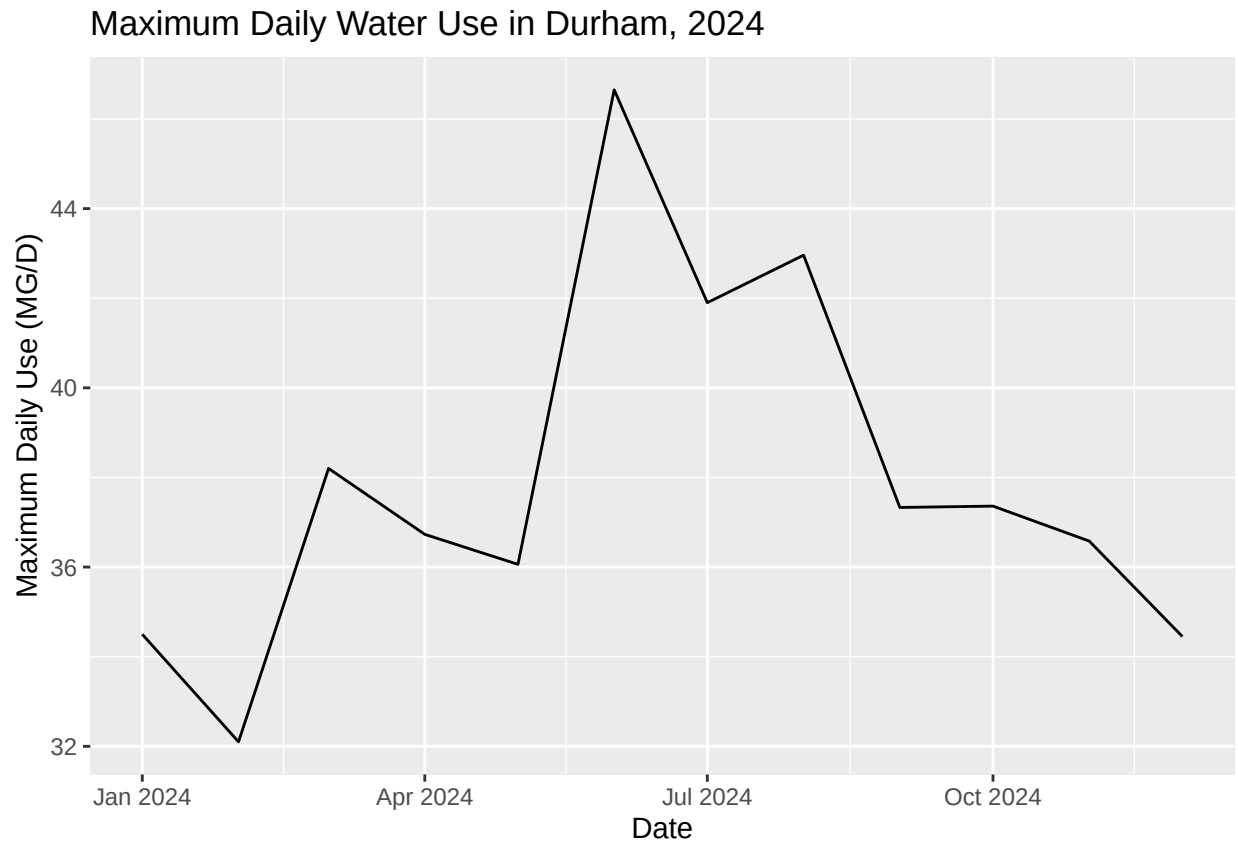
5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```

#4
Durham_water_df <- data.frame(
  "Month" = c("Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov",
             "Apr", "Aug", "Dec"),
  "MaxDayUse_MGD" = as.numeric(MaxDayUse)) %>%
mutate(Date = my(paste(Month, "2024")),
       System_Name = !!system,
       PWSID = !!PWSID,
       Ownership = !!ownership) %>%
  arrange(Date)

#5
ggplot(Durham_water_df, aes(x = Date, y = MaxDayUse_MGD)) +
  geom_line() +
  labs(
    y = "Maximum Daily Use (MG/D)",
    title = "Maximum Daily Water Use in Durham, 2024"
  )

```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)

- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.
LWSP_scraper <- function(the_year, the_PWSID){

  the_URL <- paste0(
    'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=',
    the_PWSID,
    '&year=',
    the_year)

  the_website <- read_html(the_URL)

  system_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
  ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
  MaxDayUse_tag <- "th~ td+ td"

  the_system <- the_website %>%
    html_nodes(system_tag) %>%
    html_text()

  the_ownership <- the_website %>%
    html_nodes(ownership_tag) %>%
    html_text()

  the_MaxDayUse <- the_website %>%
    html_nodes(MaxDayUse_tag) %>%
    html_text()

  MaxUse_df <- data.frame(
    "Month" = c("Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov",
               "Apr", "Aug", "Dec"),
    "MaxDayUse_MGD" = as.numeric(the_MaxDayUse)) %>%
    mutate(Date = my(paste(Month, the_year)),
           Year = !!the_year,
           System_Name = !!the_system,
           PWSID = !!the_PWSID,
           Ownership = !!the_ownership) %>%
    arrange(Date)

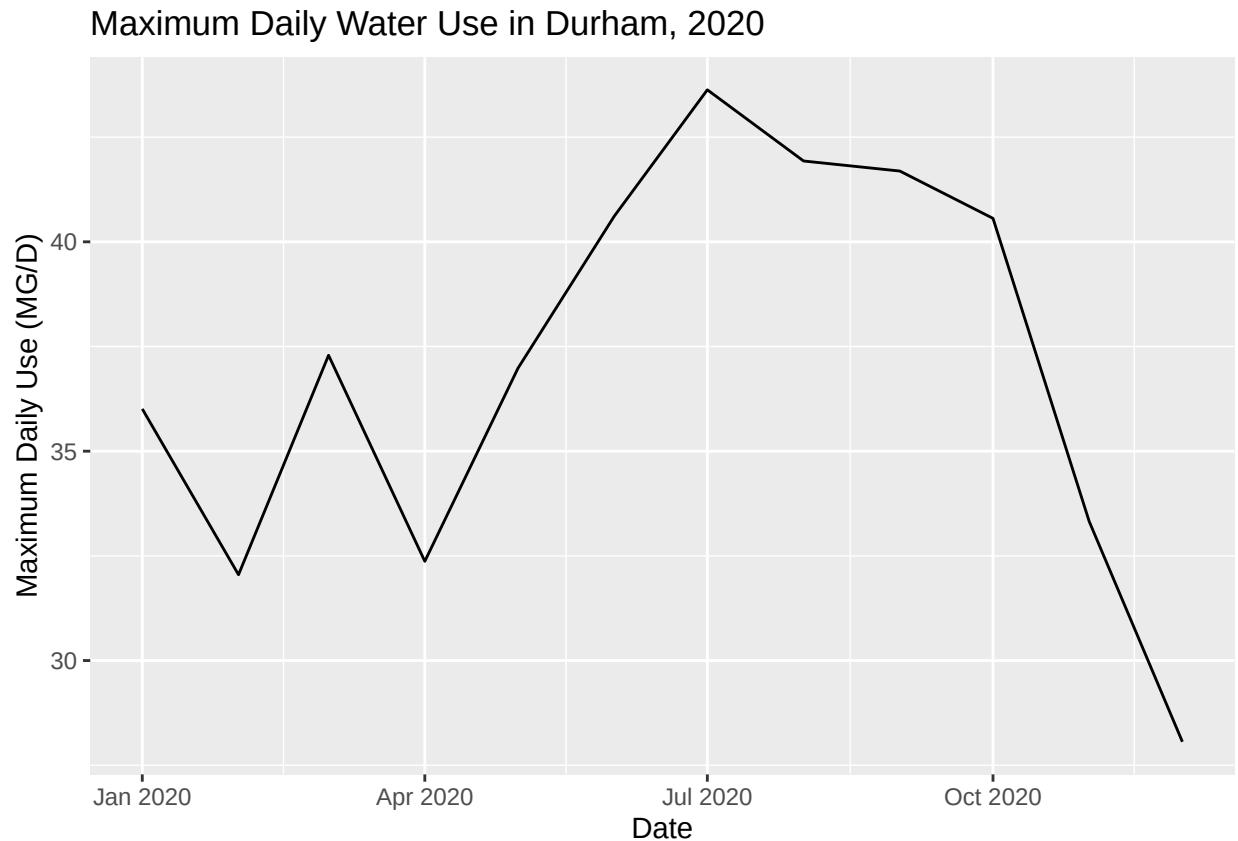
  MaxUse_df
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
Durham_2020 <- LWSP_scraper(2020, "03-32-010")

ggplot(Durham_2020, aes(x = Date, y = MaxDayUse_MGD)) +
```

```
geom_line() +
labs(
  x = "Date",
  y = "Maximum Daily Use (MG/D)",
  title = "Maximum Daily Water Use in Durham, 2020"
)
```



```
str(Durham_2020)
```

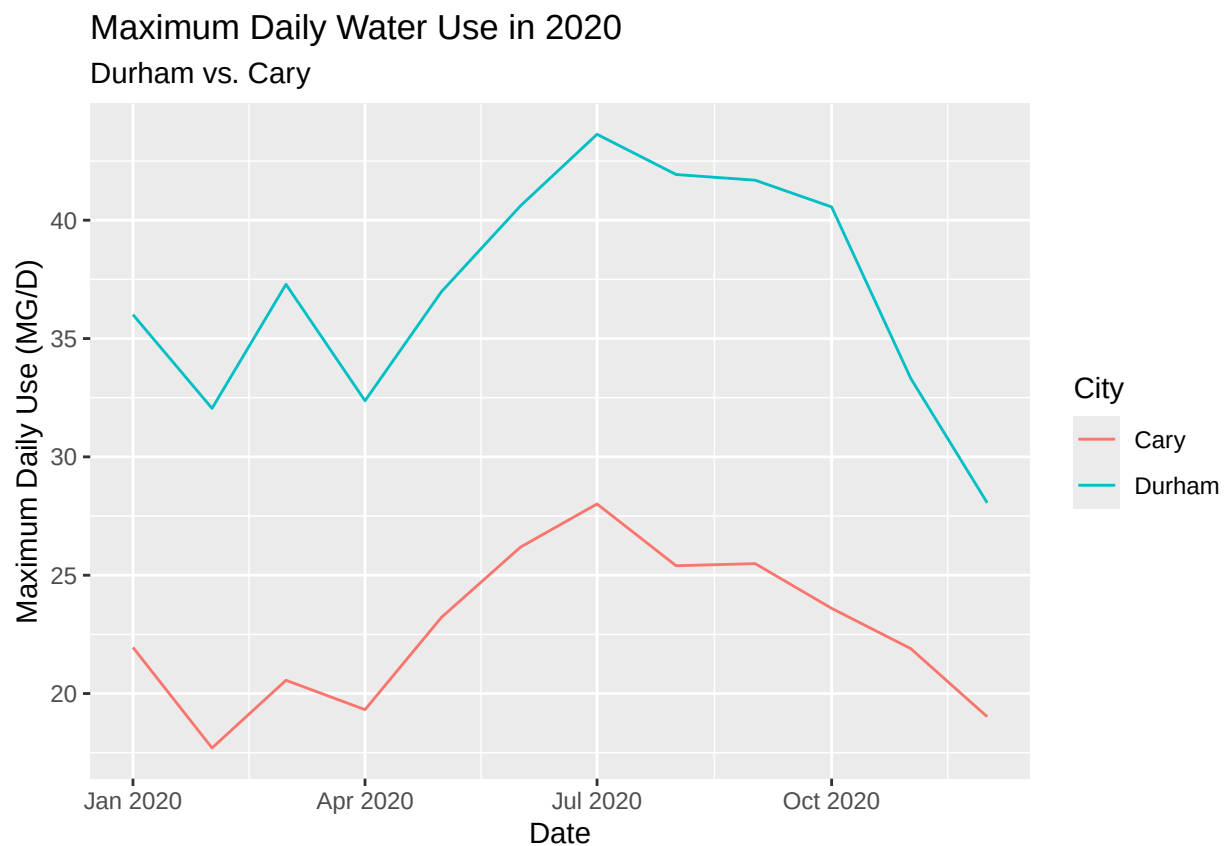
```
## 'data.frame':  12 obs. of  7 variables:
## $ Month      : chr  "Jan" "Feb" "Mar" "Apr" ...
## $ MaxDayUse_MGD: num  36 32 37.3 32.4 37 ...
## $ Date       : Date, format: "2020-01-01" "2020-02-01" ...
## $ Year       : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ System_Name : chr  "Durham" "Durham" "Durham" "Durham" ...
## $ PWSID      : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Ownership   : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
```

- Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
Cary_2020 <- LWSP_scraper(2020,"03-92-020")

Durham_and_Cary_2020 <- rbind(Durham_2020, Cary_2020)

ggplot(Durham_and_Cary_2020, aes(
  x = Date,
  y = MaxDayUse_MGD,
  color = System_Name)) +
  geom_line() +
  labs(
    y = "Maximum Daily Use (MG/D)",
    title = "Maximum Daily Water Use in 2020",
    subtitle = "Durham vs. Cary",
    color = "City"
  )
)
```



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

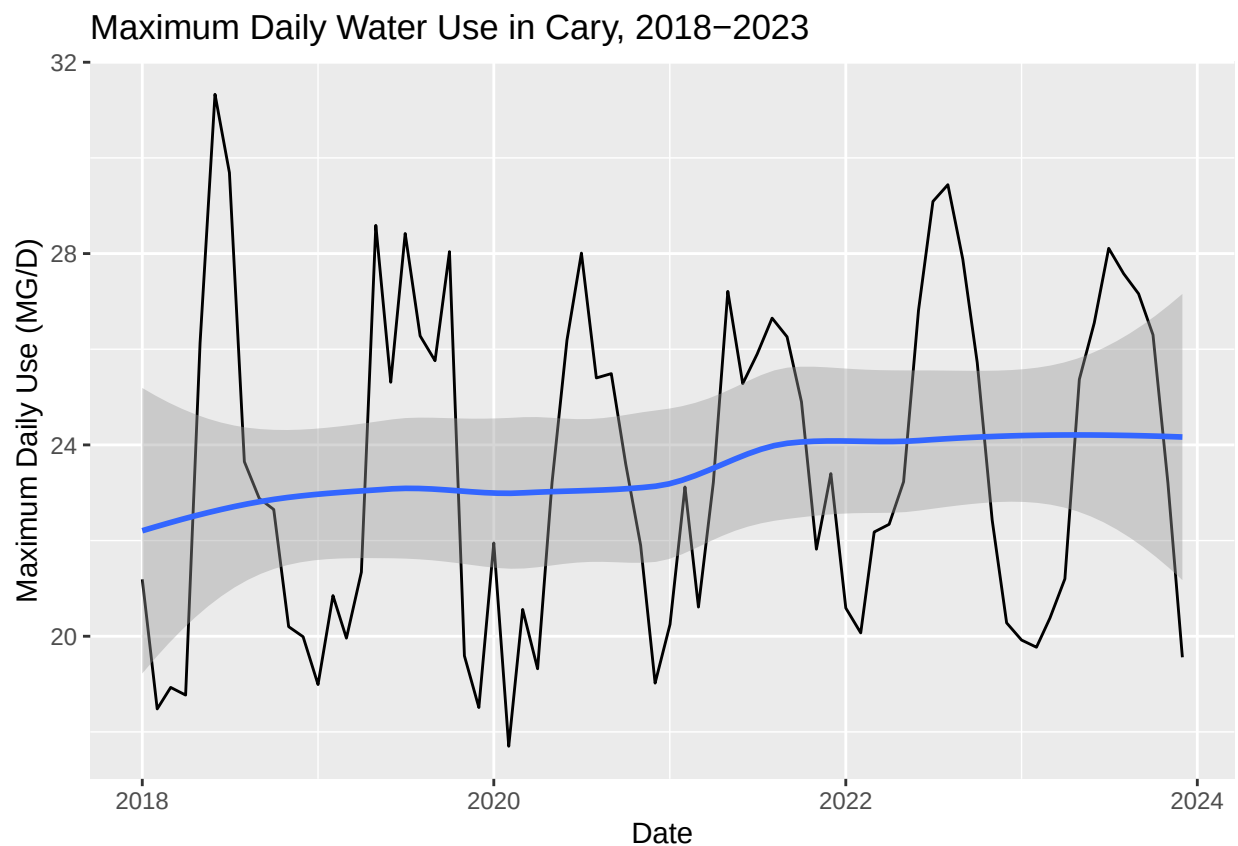
TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
Cary_2018to2023 <- map2(c(2018:2023), rep("03-92-020",6), LWSP_scraper) %>%
  bind_rows()

ggplot(Cary_2018to2023, aes(x = Date, y = MaxDayUse_MGD)) +
  geom_line() +
  geom_smooth(method = "loess") +
  labs(
    y = "Maximum Daily Use (MG/D)",
    title = "Maximum Daily Water Use in Cary, 2018-2023"
  )

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: > According to the LOESS regression line, Cary appears to have a slight upward trend in maximum water usage over time.